ZZ-M-42

APRIL 10, 1947

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

MATS, FLOOR; DENTAL-CHAIR, RUBBER

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than June 15, 1947 for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATION

A–1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification:

ZZ–R–601—Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).

B. GRADES

B–1. Dental-chair floor mats in accordance with this specification shall be of but one grade.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The mats shall be made of a rubber compound which may be natural rubber, synthetic rubber, or mixtures of the two.

C–2. *Workmanship.*—Workmanship shall be first class in every respect. Mats shall be free from blisters, cracks, imbedded foreign matter, or other physical defects which detract from their appearance or which may impair their serviceability. Mats shall be free from objectionable odors.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Design and construction.*—The mats shall be semicircular in shape with a half circle cut-out to fit against the base of a dental chair having a diameter of 24 inches. The base of the mat shall be of sponge rubber and the upper surface shall be of rubber tile that has been properly cemented and vulcanized to the base.

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E–2. *Surfaces.*—The wearing surface shall be smooth good-quality rubber tile which can be easily cleaned and is not adversely affected by floor waxes. The edges of the mat, with exception of the cut-out, shall be beveled approximately 1 inch. The underneath surface of the mat shall be treated so as to prevent sliding or creeping on a smooth floor.

E–3. *Dimensions.*—Over-all measurements shall be not less than 69 by 35 inches. Distance from inner edge of cut-out to outer edge of mat shall be 23½ inches ± ½ inch. Thickness of the tile surface shall be not less than 3/32 inch. Combined thickness of the base and tile surface shall be not less than ½ inch.

E–4. *Weight.*—Each mat shall weigh not less than 18 pounds.

E–5. *Color.*—The rubber-tile surface shall have a black background with white and green mottling.

E–6. *Flexibility.*—The mat shall not crack or show any indication of weakness when tested in accordance with paragraph F–2a.

E–7. *Friction.*—The adhesion of the tile surface to the sponge-rubber base shall be such that the rate of separation shall be not greater than 1 inch per minute when tested in accordance with paragraph F–2b.

E–8. *Modulus at 10 percent elongation.*—When tested as described in paragraph F–2c, the stress at 10 percent elongation shall be not less than 125 pounds per square inch.

E–9. *Marking.*—The mats shall be permanently and legibly marked with the name or trade-mark of the manufacturer.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—One mat shall be taken at random from each 300 or less delivered, for inspection and test.

F–2. *Inspection and tests.*—Unless otherwise specified, inspection and tests shall be in accordance with the applicable methods of Federal Specification ZZ–R–601, and as specified hereinafter.

F–2a. *Flexibility.*—Double and press the mat flat on itself in any direction for 5 minutes with the tile surface outside. Apply the test both before and after the sample has been subjected to a temperature of 158° F. for 7 days.

F–2b. *Friction.*—Apply a weight of 3 pounds to a test strip 1 inch wide and test by the dead weight method (strip specimens) as described in Federal Specification ZZ–R–601.

F–2c. *Modulus.*—Test pieces of the surface tile shall be prepared in the form of strips 1 inch wide, cut with the grain, with the sponge rubber removed, and buffed over a gage length of 5 inches. The test pieces shall be stretched at a machine speed of 6 inches per minute. The tension in pounds on the test piece at 10 percent elongation divided by the cross section of test piece in square inches represents the modulus in pounds per square inch. The reported modulus shall be the average of three determinations.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification.

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G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or order.

G–3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name of the item, the carriers' freight classification description "Mats, rubber", and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following departmental specifications and publication, of the issues in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments.

H–2. *Army and Navy.*—

H–2a. *Applicable specifications and other publication.*—

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (For Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (For Domestic Shipment).

Joint Army-Navy Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–107—Packaging and Packing for Overseas Shipment—Boxes, Wood, Wire-Bound.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100–2—Standard Specification for Marking Shipments By Contractors.[2]

Navy Department Specification:

General Specifications for Inspection of Material.[2]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[2]

[1] Applicable to Army purchases only.
[2] Applicable to Navy purchases only.

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H–2a(1). Copies of Federal and Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department. Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Federal, Joint Army-Navy, and U. S. Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–2a(2). Copies of Federal and Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–2b. *Packaging.*—Each mat shall be completely wrapped in kraft paper of not less than 30 pounds basis weight (500 sheets, 24 x 36 inches), or equal, securely taped closed with kraft gummed paper tape; each unit shall be further packaged in a fiberboard carton of appropriate size and design, conforming to the requirements of either Federal Specification LLL–B–631 or LLL–B–636, except that the material shall have a minimum bursting strength of not less than 200 pounds per square inch.

H–2c. *Packing.*—

H–2c(1). *For domestic shipment.*—Four units packaged as specified in paragraph H–2b shall be packed (for type 2 load) in an exterior container conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, or JAN–P–107 (except that the requirements for container manufacturer's identification marking shall not apply). Closure and strapping shall conform to the requirements of the appendix to the applicable box specification.

H–2c(2). *For overseas shipment.*—Four units shall be packed as specified in paragraph H–2c(1). In addition, each box shall be lined with a sealed waterproof barrier bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN–P–125 for case liners; seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140.

H–2d. *Marking.*—

H–2d(1). *Interior carton.*—Each carton and barrier shall be marked with stock numbers, nomenclature, unit, quantity, and name of manufacturer. No marking other than that required by the specification shall appear.

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H–2d(2). *Shipping containers.*—

H–2d(2)a. *Army.*—Shipping containers shall be marked in accordance with the requirements of U. S. Army Specification 100–2, and shall bear the Joint Army-Navy stock number and the Army stock number.

H–2d(2)b. *Navy.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook, and shall bear the Joint Army-Navy stock number.

I. NOTES

I–1. Purchasers should exercise any desired options offered herein. (See pars. F–2, G–1, G–2, and G–3.)

I–2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–3. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I–4. An Index of Federal Specifications may be purchased as noted in paragraph below, price to be obtained from the Superintendent of Documents.

I–5. Copies of this specification and ZZ–R–601 may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price of this specification, 5 cents; ZZ–R–601, 15 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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